

Four Months' Notice

Turning El Niño into a dengue early-warning alarm for ASEAN

A freely available climate index gives ASEAN health offices months of warning before a dengue season turns severe. We turn that lag into a two-condition alert rule any provincial health office can run today — no model, no code, no paid data.

SDG 3 Health

SDG 13 Climate

SDG 6 Water

Team [NAMA TIM] | [NAMA INSTITUSI] | Indonesia

THE PROBLEM

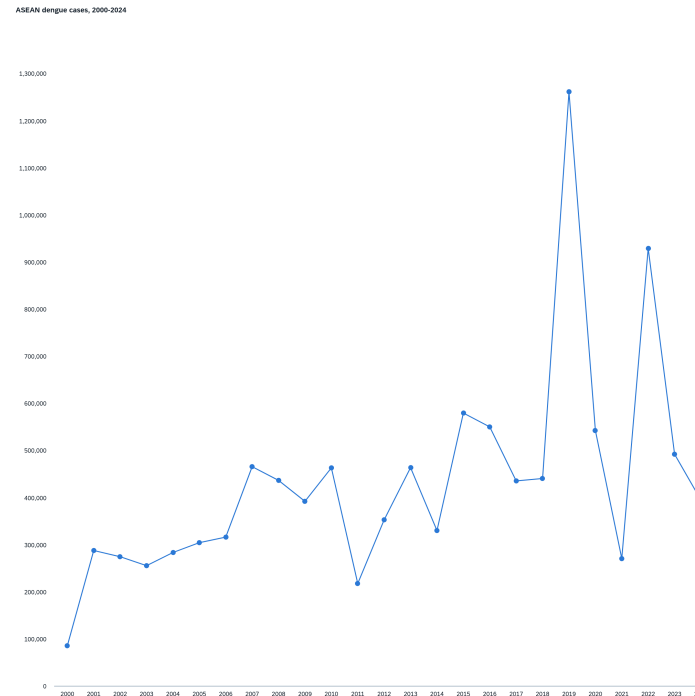
Dengue is ASEAN's fastest-moving climate-sensitive disease

1.26M

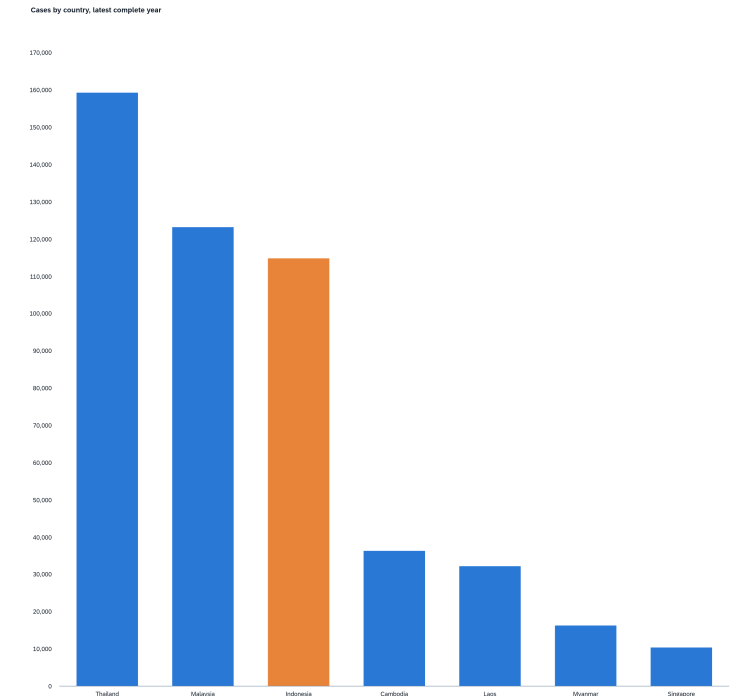
cases in 2019 — the worst year in our nine-country panel

- Vietnam 368k · Philippines 220k · Thailand 159k (latest complete year per country)
- The 2023–24 El Niño added an estimated **9.6 million cases worldwide** (Tian et al., 2025)
- Outbreaks are still fought **reactively** — after cases surge

ASEAN dengue cases, 2000–2024



Cases by country, latest complete year



THE QUESTION

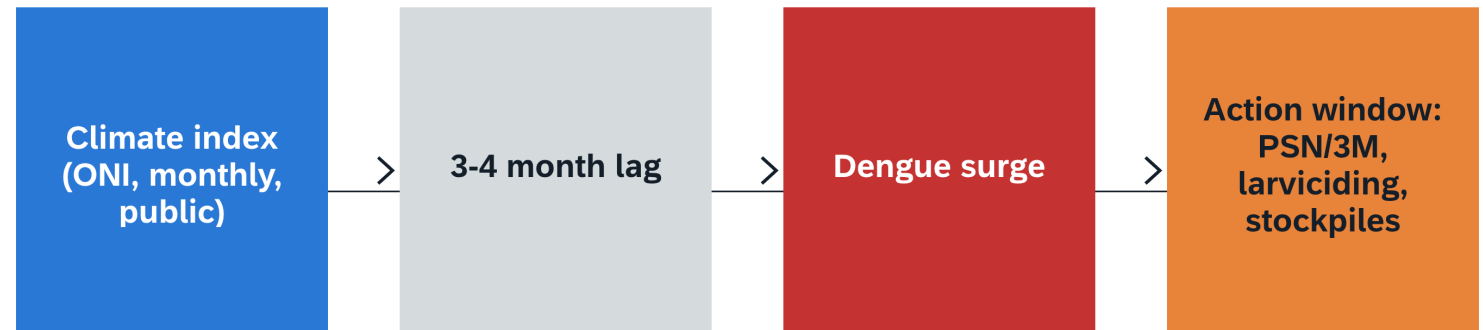
Climate data arrives months before patients do

Can a freely available climate index tell a health office how bad the coming season will be — early enough to act?

- **SDG 3** Health — reduce dengue burden
- **SDG 13** Climate — use climate information for adaptation
- **SDG 6** Water — target the most vulnerable districts

Our test: does the signal survive when we remove the seasonal cycle, hold out unseen years, and use the WHO outbreak definition?

From public index to public action



DATA & METHOD

Nine open datasets, one monthly panel, zero paid sources

DATA	ROWS	COVERAGE
Dengue cases (OpenDengue V1.3)	70,397	10 countries, 1955–2025
Dengue × climate monthly panel	55,329	province-month, lags pre-computed
Temperature & rainfall (CCKP, ERA5)	9,000	monthly, 1950–2024
ONI El Niño index (NOAA CPC)	918	monthly, 1950–2026
Water & sanitation (WHO/UNICEF JMP)	8,692	2000–2024
Population (BPS Indonesia)	140	provinces, denominators
Flood exposure (GHSL)	30,348	836 cities, 1975–2030

SDG 3

SDG 13

SDG 6

Every table joins on the same two keys — iso3 + year (plus period for the monthly dengue × climate panel). Nothing is entered by hand.

METHOD

- Monthly national series, 2010+
- Lagged correlations, 0–4 months
- **Anomalies** — seasonal cycle removed before we believe any correlation
- A two-condition trigger rule
- Out-of-sample test on unseen years
- WHO/PAHO endemic-channel benchmark

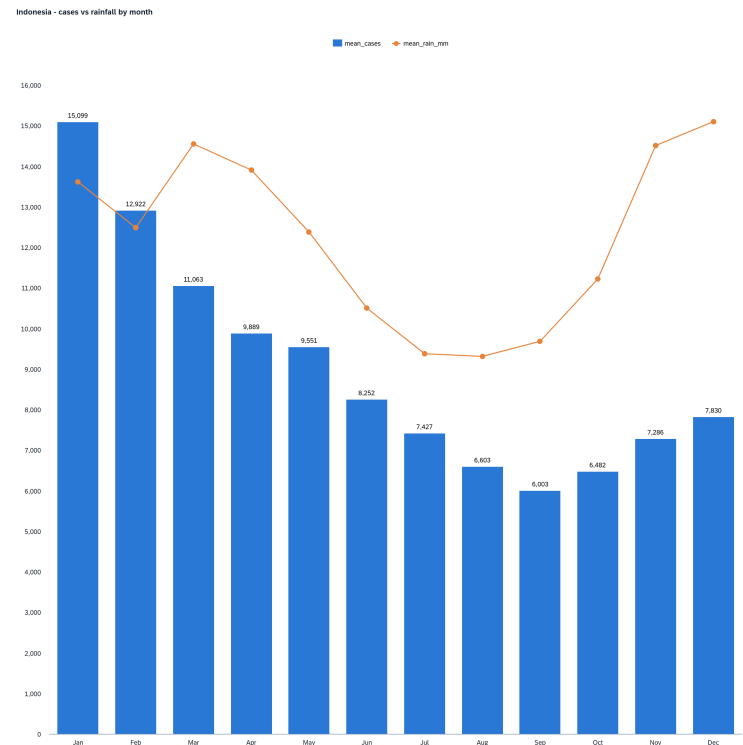
All charts built in SAP Analytics Cloud.

ANALYSIS 1 — THE SEASON

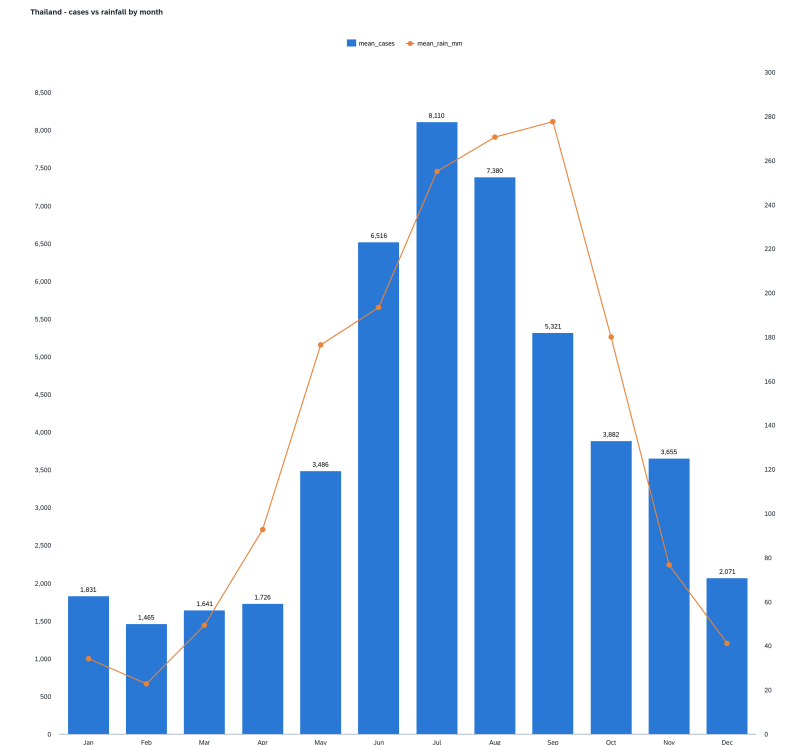
Dengue has a calendar — and it is not the same calendar everywhere

- **Indonesia:** cases peak **Jan–Apr** (Jan avg 15,099/month vs Sep 6,003 — $2.5\times$), right after the December rainfall peak
- **Thailand:** cases climb with the **onset** of the wet season and peak **Jun–Sep** (Jul avg 8,110 vs Feb 1,465 — $5.5\times$)
- Same disease, opposite halves of the year → a regional alarm must be **local in timing**

Indonesia — cases vs rainfall by month



Thailand — cases vs rainfall by month



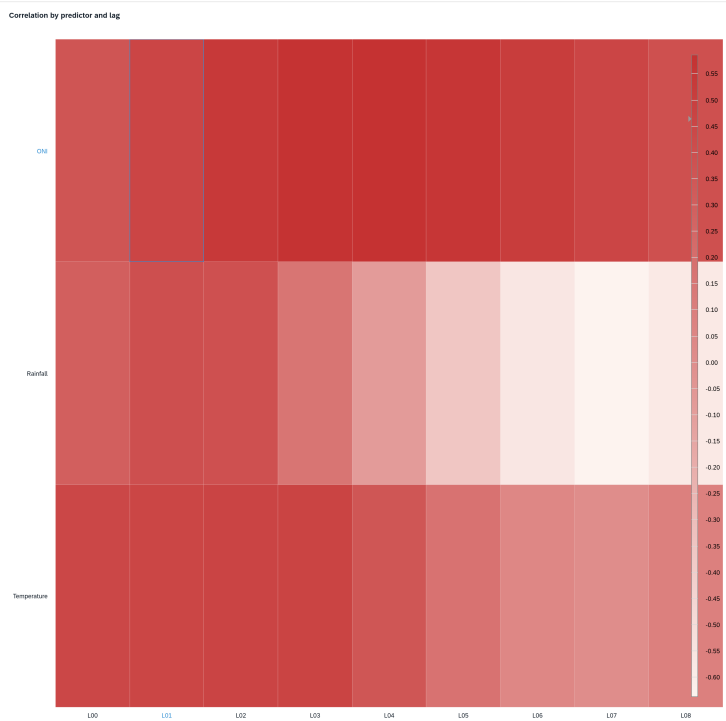
ANALYSIS 2 — THE LAG

The climate signal arrives 1–4 months before the cases

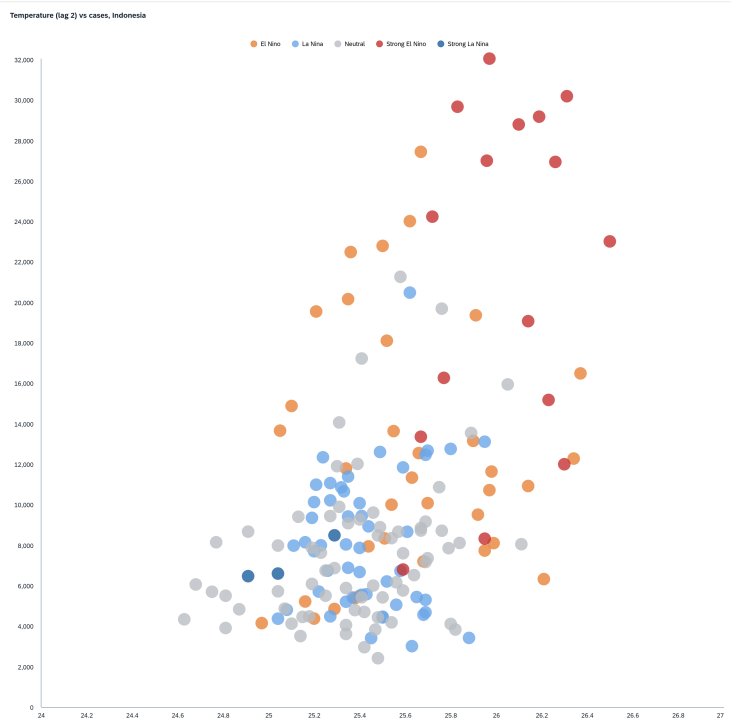
COUNTRY	BEST PREDICTOR	LAG	R
Indonesia	ONI (El Niño)	4 mo	+0.59
Indonesia	Temperature	3 mo	+0.47
Indonesia	Rainfall	1 mo	+0.40
Thailand	Temperature	3 mo	+0.57
Thailand	Rainfall	same mo	+0.48

The lag *is* the opportunity: it is the time a health office has to act.

Correlation by predictor and lag



Temperature (lag 2) vs cases, Indonesia



ANALYSIS 3 — THE ACE CARD

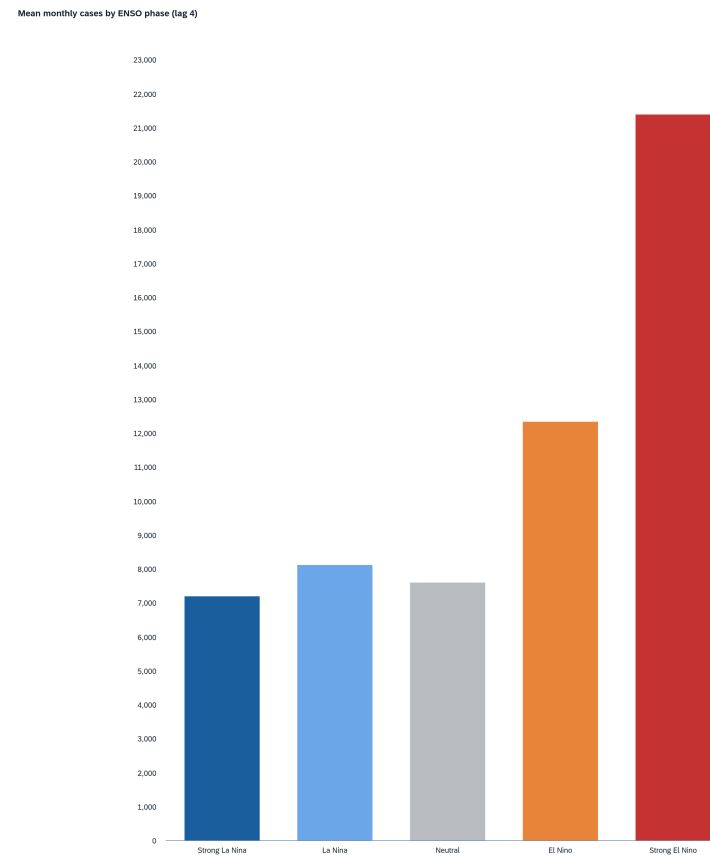
A strong El Niño nearly triples Indonesia's monthly dengue burden

2.8×

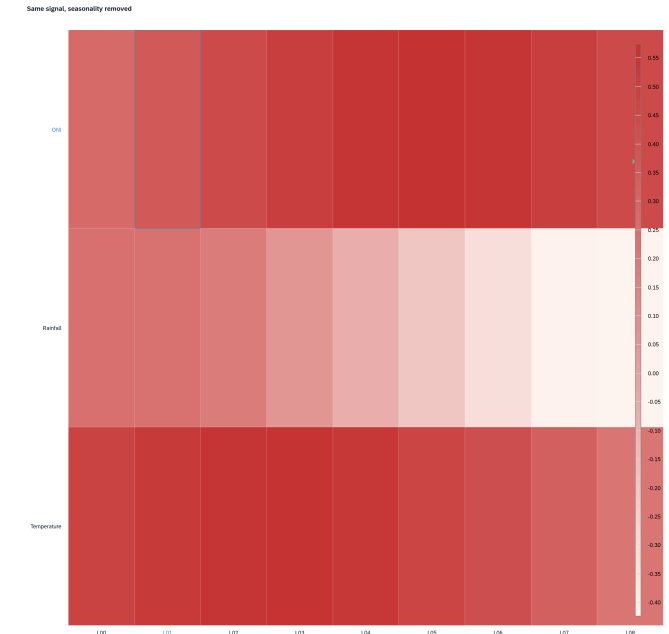
21,383 vs 7,597 cases per month, grouped by the ENSO phase four months earlier

- Grouped by the phase **four months earlier** — so this is a forecast, not hindsight
- **Not just seasonality**: after removing the seasonal cycle the ONI correlation holds at **$r = +0.56$**
- **Not an aggregation artefact**: cases are higher in El Niño months in **33 of 34 Indonesian provinces** (sign test $p = 2 \times 10^{-9}$) and 70 of 77 Thai provinces
- Rainfall does *not* survive the same test ($+0.40 \rightarrow +0.24$) — we report that honestly

Mean monthly cases by ENSO phase (lag 4)



Same signal, seasonality removed

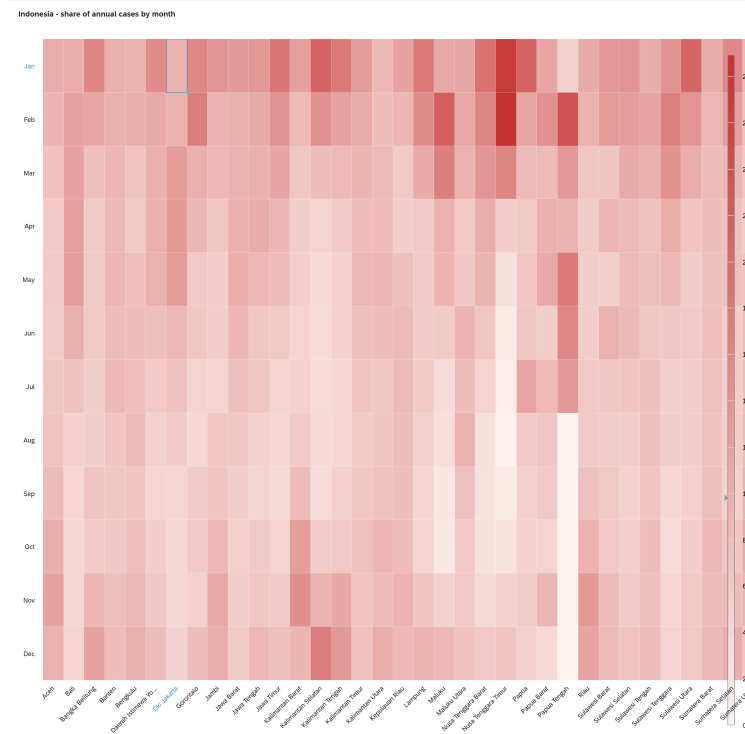


ANALYSIS 4 — WHERE AND WHEN

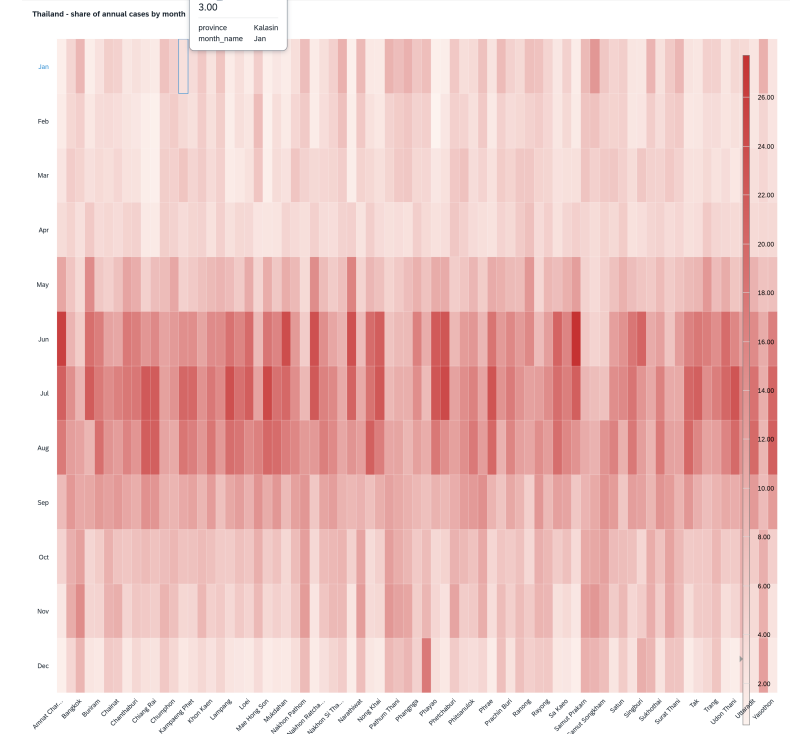
The alarm must be set province by province

- **Indonesia:** 20 of 35 provinces peak in **January**, 8 in February — but **Bali peaks in May** and North Kalimantan in December
- **Thailand:** 38 of 77 provinces peak in **July**, 16 in August
- A single national alert date would be wrong for a third of Indonesia

Indonesia — share of annual cases by month



Thailand — share of annual cases by month

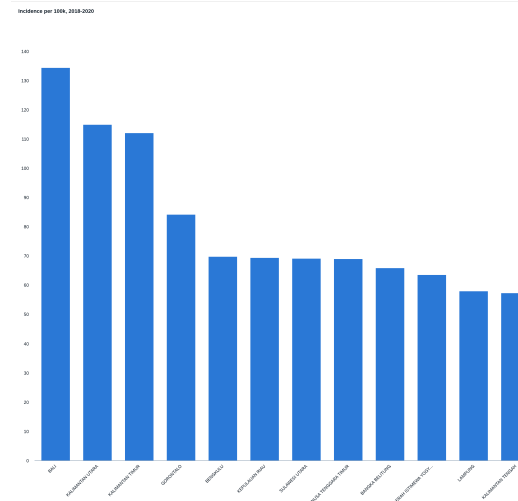


ANALYSIS 5 — WHO GETS HIT HARDEST

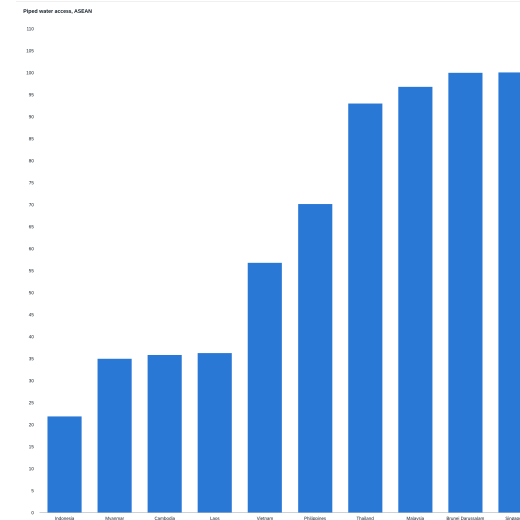
The same alarm, aimed at the most exposed places

- Highest incidence 2018–2020: **Bali 134**, North Kalimantan 115, East Kalimantan 112 per 100k — **14× the lowest province** (Papua, 9)
- Indonesia has the **lowest piped-water access in ASEAN: 21.8%** (JMP 2024) — households store water, and stored water breeds *Aedes*
- **12.9 million people in Bangkok** and 1.7 million in Jakarta live in the 1-in-100-year flood zone
- We use this to **target** the alarm, not to explain it (see p. 14)

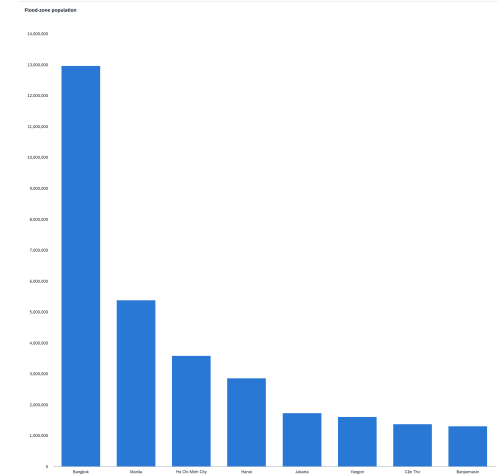
Incidence per 100k, 2018–2020



Piped water access, ASEAN



Flood-zone population

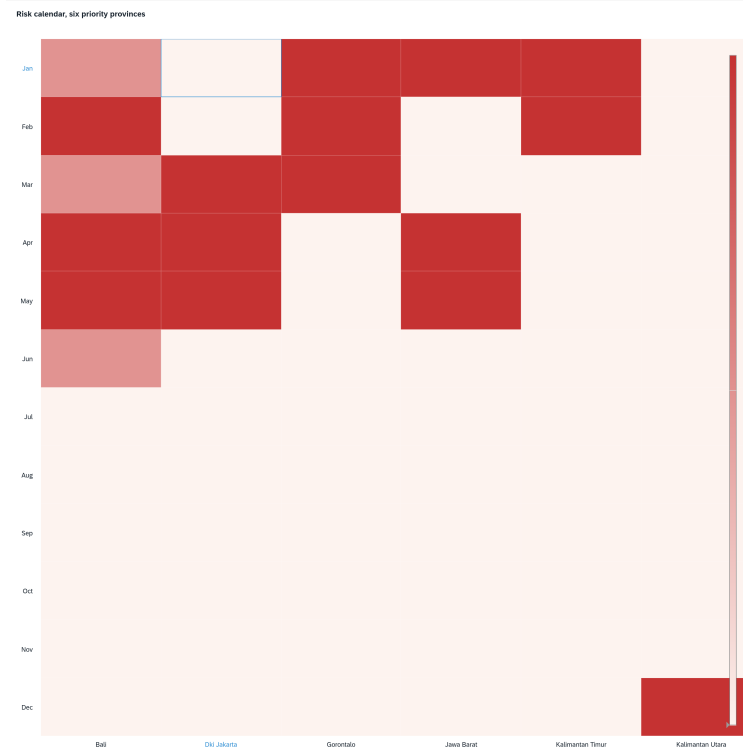


SYNTHESIS

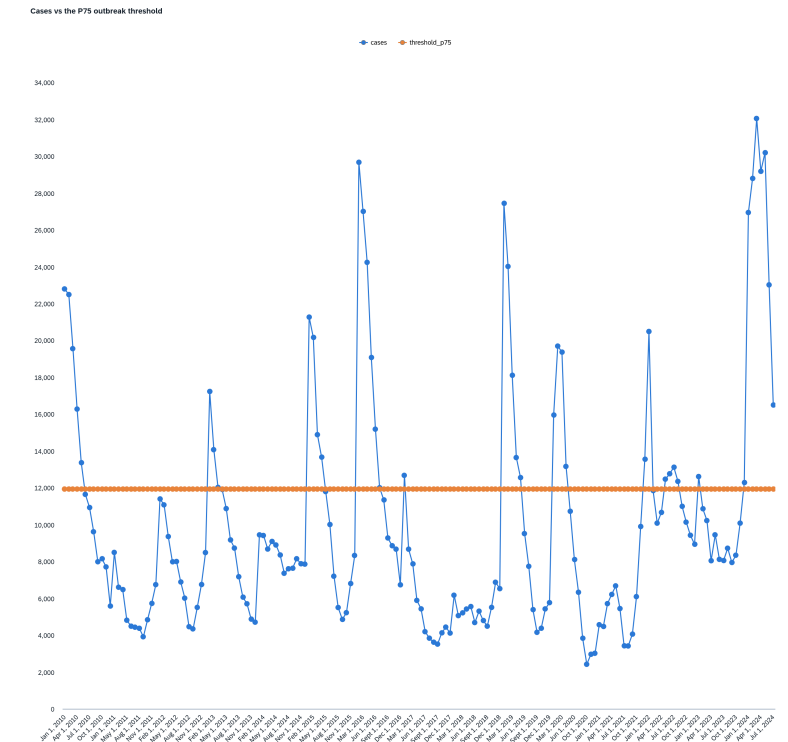
From signal to schedule: a risk calendar every health office can read

- Each province gets three states — **Peak** · **Watch** · **Normal** — from its own case history
- The El Niño index sets **how severe** the coming peak will be; the calendar sets **when** it arrives
- Real lead times, every El Niño episode since 2009: **0–3 months** on observed ONI — and longer on NOAA/BMKG *forecast* ONI

Risk calendar, six priority provinces



Cases vs the P75 outbreak threshold



THE SOLUTION

Two conditions. No model. No code.

“If the month is Jan–Apr AND ONI four months ago was $\geq 0.5 \rightarrow$ raise alert”

11.6%

of months fire the alert

100%

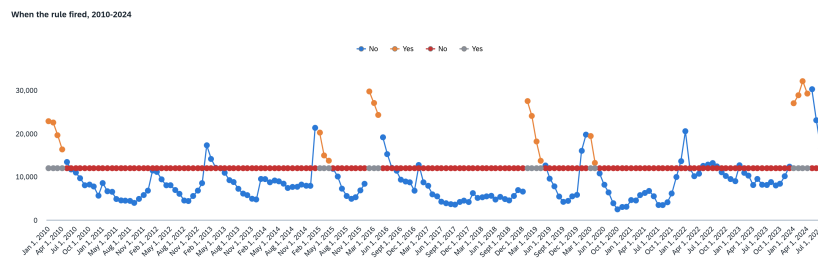
precision — every alert was a top-quartile dengue month

4×

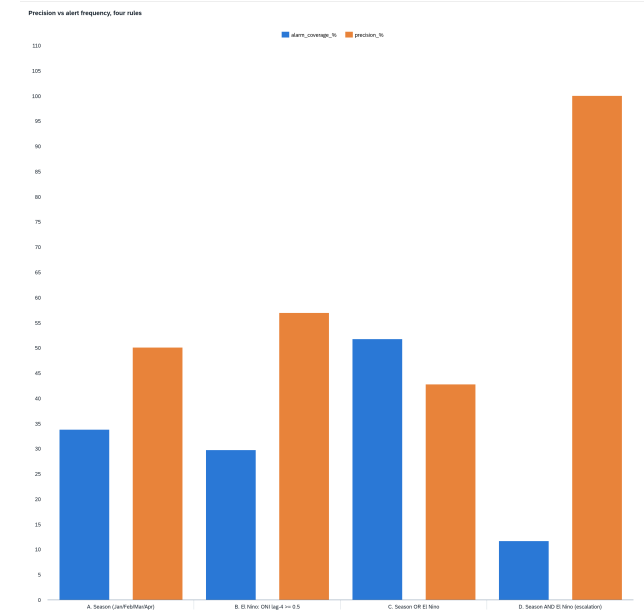
lift over the base rate

- Both inputs are public and free: NOAA publishes ONI monthly, the calendar comes from the office's own records
- Runs in a spreadsheet, a WhatsApp broadcast, or this SAC dashboard — competitors promise machine learning; we promise something a provincial health office can start next week

When the rule fired, 2010–2024



Precision vs alert frequency, four rules

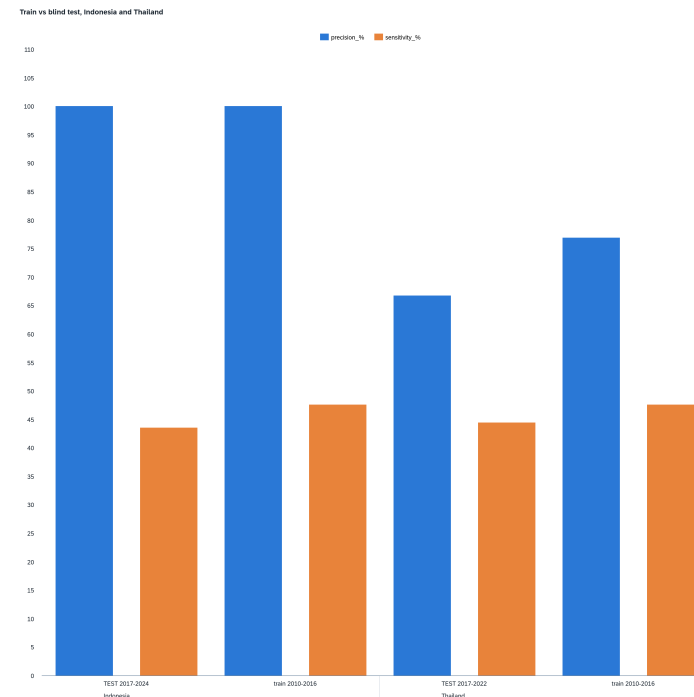


VIABILITY

We tried to break our own rule. It held.

- **Out-of-sample test:** rule built on 2010–2016, tested blind on 2017–2024 → **precision still 100%**, sensitivity 44%, lift 3.9
- Against the **WHO/PAHO endemic-channel** definition — which removes seasonality by construction — specificity **91%**, lift 2.5, and *all* of the value comes from the El Niño condition, not the season
- Published dengue EWS report PPV 43–86% (17-system review); we reach a comparable PPV **with no statistical model and monthly data**
- Replicates in Thailand with its own best predictor (temperature anomaly, lag 2): lift 2.7

Train vs blind test, Indonesia and Thailand



How we compare to published EWS

How we compare to published EWS

[system](#) | [1](#)

Measures		sensitivity_low	sensitivity_high	precision_low	precision_high
system	source				
17 published EWS (range)	Baharom et al. 2022	50	100	43	86
EWARS-csd, Colombia	Schlesinger et al. 2024	97	97	75	75
Rule D vs P75 (blind test)	this notebook	44	44	100	100
Rule D vs endemic channel	this notebook	29	29	30	30
WHO-TDR EWARS, Mexico	Cardenas et al. 2021	100	100	83	83

ADOPTION IN THREE PHASES

- 1 Bali, N. & E. Kalimantan pilot →
- 2 all 38 provinces →
- 3 ASEAN peers

Cost: open data, monthly refresh, existing staff.

IMPACT — INDONESIA ALONE

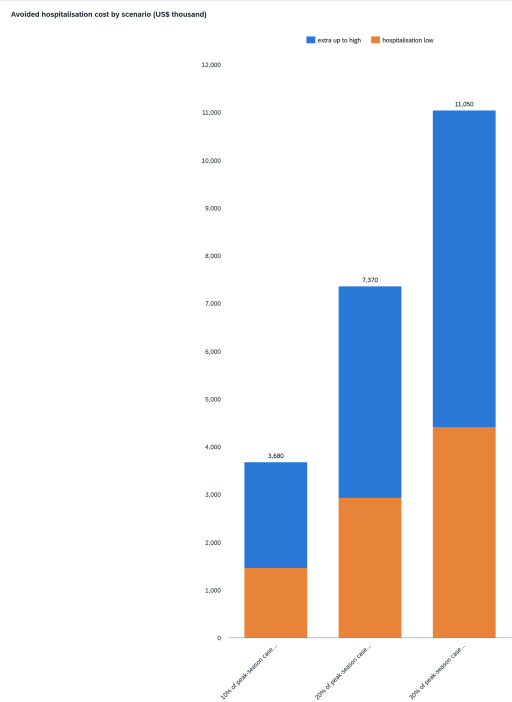
Acting 4–8 weeks earlier is worth US\$1.5–3.7 million a year

4,656

cases/year prevented in Indonesia, under the most conservative scenario

- Indonesia averages **104,753 cases/year** (complete years 2015–2023); **44% fall in Jan–Apr**
- Prevent just **10%** of peak-season cases (an assumption, not a finding) = 4,656 cases/year
- At Indonesia’s published hospitalisation cost of US\$316–791/case → **US\$1.5–3.7 M (Rp 24–59 billion)/year**
- For scale: dengue costs Indonesia **US\$381–681 M/year** (Nadjib 2019; Wilastonegoro 2020)
- Lead time is real: in the 2023–24 episode the alarm fired **Sep 2023**; cases crossed the peak threshold **Dec 2023**

Avoided hospitalisation cost by scenario



Lead time, every El Niño episode since 2009

Lead time, every El Nino episode since 2009

episode episode lead_time_months

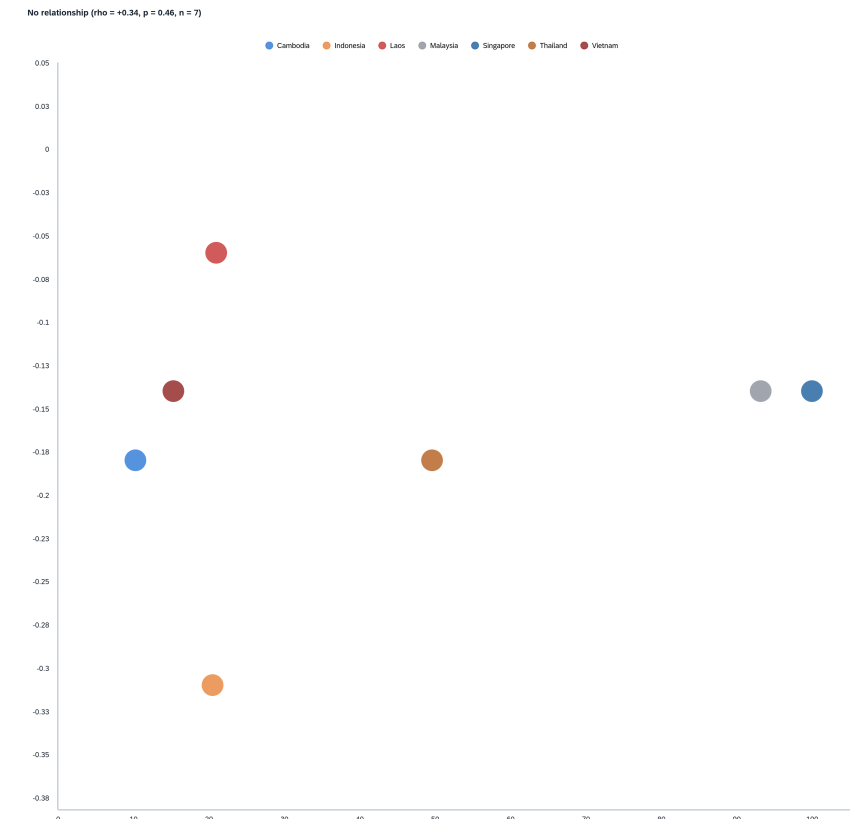
episode	alarm	cases_cross	lead_time_months
2009/10	Dec 1, 2009 (2009)	Jan 1, 2010 (2010)	1
2014/16	Feb 1, 2015 (2015)	Feb 1, 2015 (2015)	0
2018/19	Jan 1, 2019 (2019)	Jan 1, 2019 (2019)	0
2019/20	Mar 1, 2020 (2020)	Mar 1, 2020 (2020)	0
2023/24	Sept 1, 2023 (2023)	Dec 1, 2023 (2023)	3

LIMITS & DATA ETHICS

What we tested and threw away

- **We falsified our own innovation angle.** “Drought $\times$ water access” looked promising ($r = -0.31$) but collapsed after controlling for El Niño (partial $r = -0.15$) and showed no moderation by piped access ($\rho = +0.34$, $p = 0.46$, $n = 7$). **We dropped the claim.**
- **We do not claim a 2023–24 record** — our panel is incomplete for Vietnam and the Philippines after 2022.
- **Incidence only for 2018–2020**, the only years with official BPS denominators. Elsewhere we use raw counts and never rank regions by them.
- **Flood and disaster records look correlated with dengue ($r = +0.66$) — until you divide by population ($r = -0.14$).** We report the null.
- Correlation is not causation: this is an **early-warning** claim, with mechanism cited from the literature.

No relationship ($\rho = +0.34$, $p = 0.46$, $n = 7$)



HALAMAN CADANGAN

Kuota resmi 15 halaman — satu halaman masih kosong

Halaman 1–14 sudah memakai 14 dari 15 halaman yang diizinkan (halaman Referensi tidak dihitung). **Satu halaman masih tersisa.**

Dari 15 dek finalis ASEAN DSE 2024–2025 yang dianalisa, **seluruhnya** menutup bagian solusi dengan sebuah **produk bernama beserta mockup antarmuka** — CircularEat, Agrigrow, MentaLink, GreenLoop, Nurture Hub. Panitia sendiri menyatakan tim “encouraged to create an app or technological prototype” untuk memperkuat pencalonan di National Finals.

Storyboard ini berhenti pada *aturan*, bukan *produk*. Halaman ini adalah tempatnya: satu nama, satu mockup dashboard SAC atau aplikasi, dan siapa yang menekan tombolnya.

Isi halaman ini, atau hapus seluruh blok halaman ini di berkas .tex — dek tetap sah dengan 14 halaman.

REFERENCES

Data sources and literature

DATA

OpenDengue V1.3 (doi:10.6084/m9.figshare.24259573.v4) · World Bank Climate Change Knowledge Portal (ERA5 0.25°) · NOAA CPC Oceanic Niño Index · WHO/UNICEF Joint Monitoring Programme 2025 · BPS Indonesia (WebAPI) · Global Human Settlement Layer (JRC) · SIPSN KLHK · BNPB.

LITERATURE

Tian et al. 2025 *Nat Commun* · Djaafara et al. 2026 *PLOS NTD* · Sutriyawan et al. 2025 · Lowe et al. 2021 *Lancet Planet Health* · Gibb et al. 2023 *Nat Commun* · Baharom et al. 2022 · Cárdenas et al. 2021 · Schlesinger et al. 2024 · Hussain-Alkhateeb et al. 2018 (WHO-TDR EWARS) · Nadjib et al. 2019 *PLoS NTD* · Wilastonegoro et al. 2020 *AJTMH* · Sasmono et al. 2026 *IJID Regions* · Childs et al. 2025 *PNAS*.

TODO sebelum submit: ganti seluruh pranala consensus.app di `analisis/keluaran/referensi_per_klaim.md` dengan DOI jurnal, lalu lengkapi sitasi di halaman ini menjadi format penuh (penulis, tahun, judul, jurnal, DOI). Halaman referensi tidak dihitung dalam batas 15 halaman — boleh lebih dari satu halaman.